

Bond-disordered non-collinear magnets: Stability of the goldstone modes

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Much like temperature, in some systems quenched disorder prevents spontaneous symmetry breaking below a certain lower critical dimension [?]. For continuous symmetries, break down of a long ranged order is signalled by the infrared catastrophe of the Goldstone mode fluctuations of the ordered state. This argument, attributed to Mermin and Wagner [?], has a straightforward parallel in magnetically ordered systems coupled with random fields [?]. In the following we shall show that the random exchange couplings also become crucial to the fate of a certain class of magnetic order, the non-collinear antiferromagnets.

I. THE GOLDSTONE MODE THEORY OF COLLINEAR AND NON-COLLINEAR MAGNETS

A. Goldstone modes in collinear magnets

1. The model

Massless goldstone modes are generated when the continuous symmetry of a hamiltonian is spontaneously broken. The classic example is an exchange coupled ferromagnet with a global spin rotation invariance. The continuum approximation for the hamiltonian for such a system is given by,

$$H = \frac{\rho}{2} \int d^d x (\nabla \mathbf{S}(x))^2 \quad (1)$$

where ρ is the spin-stiffness parameter. At zero temperature the ground state breaks the full $O(3)$ rotational symmetry of the model and assumes a configuration with all spins parallel to each other in a collinear pattern. The ground state therefore has a reduced $O(2)$ symmetry related to the rotation around the magnetization axis. In a generic model with a symmetry breaking from $O(N)$ to $O(N - 1)$, the elementary excitations are the goldstone modes linked with the generators of the quotient group $O(N)/O(N - 1)$, also called, the broken generators. Let us consider a symmetry broken ground state for an $O(N)$ ferromagnet, $\mathbf{S}_0 = (0 \dots 0, 1)^T$. It is immediately apparent that there is a degenerate continuum of states related to the residual $O(N - 1)$ rotational symmetry spanning the $N - 1$ dimensional subspace orthogonal to the magnetization axis.

Any symmetry operation that goes beyond this subspace would generate massless goldstone modes. We can make a particular gauge choice,

$$U = \begin{pmatrix} \sqrt{I_{N-1} - \boldsymbol{\pi}^T \boldsymbol{\pi}} & \boldsymbol{\pi} \\ -\boldsymbol{\pi}^T & \sqrt{1 - \boldsymbol{\pi} \boldsymbol{\pi}^T} \end{pmatrix} \quad (2)$$

where, $U \in O(N)/O(N-1)$ and find, $\mathbf{S} = U\mathbf{S}_0 = (\boldsymbol{\pi}, \sqrt{1 - \boldsymbol{\pi} \boldsymbol{\pi}^T})^T$. By pluggin this back to the hamiltonian we immediately arrive at the familiar $O(N)$ NL σ -model hamiltonian.

$$\begin{aligned} H &= \frac{\rho}{2} \int d^d x \left(\boldsymbol{\nabla} \pi_i \cdot \boldsymbol{\nabla} \pi_i + \frac{(\pi_i \boldsymbol{\nabla} \pi_i)^2}{1 - \pi_i \pi_i} \right) \\ &= \int d^d x g_{ij}(\boldsymbol{\pi}) \boldsymbol{\nabla} \pi_i \cdot \boldsymbol{\nabla} \pi_j \end{aligned} \quad (3)$$

For classical spins, this model is renormalizable around the lower critical dimension $d = 2 + \epsilon$ with dimensional regularization. The metric βg_{ij} effectively plays the role of an inverse temperature and the perturbative expansion can be organized as a low temperature expansion. For quantum spins the action takes up the form,

$$S = \int d\tau \int d^d x g_{ij}(\boldsymbol{\pi}) \partial_\mu \pi_i \partial^\mu \pi_j \quad (4)$$

and a similar renormalization scheme follows at $d = 1 + \epsilon$. Although, in this case the metric is understood to be the stiffness parameter for the spins.

The $O(N)$ NL σ -model is however special because although the expansion of the metric coefficient gives rise to infinite number of vertices, there is only one coupling that flows (here, ρ). Yet, NL σ -model is renormalizable at $D = d + z = 2$ and upto one-loop order one can find the renormalized value of the stiffness parameter or temperature by considering the perturbative correction to the propagator alone. Given the model is massless, in a wilsonian treatment the only non-trivial element turns out to be the field strength renormalization. A standard technique to resolve this problem is to consider a trivial renormalization of an external field coupled to the spin or alternatively ensuring that both the massless $\boldsymbol{\pi}$ goldstone modes and the massive $\sigma = \sqrt{1 - \boldsymbol{\pi} \boldsymbol{\pi}^T}$ mode have the same scaling. Both of these schemes are manifestation of the fact that one has to consider the fully symmetric model to find the scaling parameters.

2. Renormalization at $D = 2$

B. $\text{NL}\sigma$ -model in a symmetric coset space

1. The model

Although, the goldstone mode effective action of the $O(N)$ symmetric ferromagnet followed quite naturally by considering a particular gauge choice for the broken generators in the preceeding section, in the following we shall consider a more generic approach to arrive at a $\text{NL}\sigma$ -model effective action of a spontaneous symmetry breaking G/H . This will be useful for the particular case of our interest, the non-collinear antiferromagnets. In the context of the following discussion, e.g. $G/H = O(N)/O(N-1)$ shall describe what is known as a symmetric coset space but the extension to the non-collinear case will be more complicated than that.

2. The renormalization

C. $\text{NL}\sigma$ -model for non-collinear antiferromagnets

1. The model

As we saw in the preceeding section, the $O(N)/O(N-1)$ spontaneous symmetry breaking of simple ferromagnets can be understood in the context of symmetric coset space and a single coupling parameter, identified as either the spin-stiffness or the temperature depending on whether one considers quantum spins or finite temperature classical spins. The situation is slightly different with the non-collinear magnets, e.g. the triangular lattice heisenberg antiferromagnets where the original symmetry of the spins are fully broken with $O(3)/Z_2 \simeq SO(3)$ but there is still a residual C_{3v} symmetry of the lattice left in the problem. In a continuum description, this leads to a symmetry breaking pattern of $O(3) \times O(2)/O(2)$, where the $O(2)$ is not a physical symmetry of the goldstone bosons but the extension of the lattice symmetry [?]. As we show below, it is possible to arrive at this conclusion from a microscopic analysis [?].

We consider the antiferromagnetic heisenberg hamiltonian in a triangular lattice and consider an expansion around the 120° ordered state. The goldstone modes shall be excited by the generators of the broken symmetry group, $SO(3)$. Our microscopic to continuum modelling would involve considering plaquettes of elementary upward triangles, finding the low energy contributions of a departure from the 120° order and ultimately coarse-graining within each elementary triangle. We start by denoting the three sublattice orientations of the 120° order, $\mathbf{n}_A$, $\mathbf{n}_B$ and $\mathbf{n}_C = -(\mathbf{n}_A + \mathbf{n}_B)$ with a spin-excitation given by $\mathbf{S}_\alpha = U\mathbf{n}_\alpha$,

where $U \in SO(3)$. We consider the hamiltonian within an elementary triangular plaquette.

$$H = J (\mathbf{S}_A(\mathbf{x}_A) \cdot \mathbf{S}_B(\mathbf{x}_B) + \mathbf{S}_B(\mathbf{x}_B) \cdot \mathbf{S}_C(\mathbf{x}_C) + \mathbf{S}_C(\mathbf{x}_C) \cdot \mathbf{S}_A(\mathbf{x}_A)) \quad (5)$$

By performing a gradient expansion coupled with a coarse-graining of the spatial co-ordinate to capture the low energy pieces, we arrive at the following expression.

$$\begin{aligned} H &= JS^2a^2 (\mathbf{n}_AU^{-1} \cdot \nabla^2 U \mathbf{n}_B - \mathbf{n}_BU^{-1} \cdot \nabla^2 U (\mathbf{n}_A + \mathbf{n}_B) - \mathbf{n}_AU^{-1} \cdot \nabla^2 U (\mathbf{n}_A + \mathbf{n}_B)) \\ &= JS^2a^2 (\mathbf{n}_AU^{-1} \cdot \nabla^2 U \mathbf{n}_B - (\mathbf{n}_A + \mathbf{n}_B)U^{-1} \cdot \nabla^2 U (\mathbf{n}_A + \mathbf{n}_B)) \end{aligned} \quad (6)$$

The single gradient term has been dropped from the expansion as it can be easily shown to have no contribution at all from the analysis that follows. Now, a transformation to the orthonormal basis, $\mathbf{e}_1 = \sqrt{\frac{3}{2}}(\mathbf{n}_A + \mathbf{n}_B)$ and $\mathbf{e}_2 = \sqrt{\frac{1}{2}}(\mathbf{n}_A - \mathbf{n}_B)$ re-arranges the terms of the hamiltonian in a convenient form.

$$H = -\frac{JS^2a^2}{2} (\mathbf{e}_1U^{-1} \cdot \nabla^2 U \mathbf{e}_1 + \mathbf{e}_2U^{-1} \cdot \nabla^2 U \mathbf{e}_2) \quad (7)$$

An extra $O(2)$ symmetry within the $(\mathbf{e}_1, \mathbf{e}_2)$ vector space is immediately apparent. Defining $S_0 = (\mathbf{e}_1, \mathbf{e}_2)$ and performing a gauge transformation $U = US_0$ we can re-write,

$$\begin{aligned} H &= -\frac{JS^2a^2}{2} \text{Tr} (S_0^{-1}U^{-1} \nabla^2 U S_0) \\ &= -\frac{JS^2a^2}{2} \text{Tr} (U^{-1} \nabla^2 U) \end{aligned} \quad (8)$$

However, it has been shown that for a complete renormalization of the problem one has to consider a third unit vector $\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$ [?]. This introduces a second coupling to the problem with the complete hamiltonian now given by,

$$\begin{aligned} H &= p_1 (\mathbf{e}_1U^{-1} \cdot \nabla^2 U \mathbf{e}_1 + \mathbf{e}_2U^{-1} \cdot \nabla^2 U \mathbf{e}_2) + p_2 (\mathbf{e}_3U^{-1} \cdot \nabla^2 U \mathbf{e}_3) \\ &= \text{Tr} (P(U^{-1} \nabla U)^2) \end{aligned} \quad (9)$$

with gauge transformation $U = U(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ generating the full $SO(3)$ goldstone modes and the diagonal coupling matrix $P = \text{diag}(p_1, p_1, p_2)$ ensuring the $O(3) \times O(2)/O(2)$ nature of the excitations. Given $\pi_1(SO(3)) = Z_2$, the order parameter U can have Z_2 point defects which are gapped out at zero temperature. With this knowledge we can immediately write down the quantum action of the full goldstone mode

problem [? ?].

$$S = \int d\tau \int d^d x [\text{Tr} (P_\tau (U^{-1} \partial_\tau U)^2) + \text{Tr} (P_x (U^{-1} \nabla U)^2)] \quad (10)$$

where the two distinct diagonal matrices, P_τ and P_x presupposes the existence of three different spin-wave velocities of the problem as obtained from a spin-wave expansion around the ordered state.

2. Multi-parameter RG

II. BOND-DISORDERED NON-COLLINEAR ANTIFERROMAGNET

1. Random field systems and the replica approach

2. Goldstone bosons and bond disorder

In our discussion, we closely follow the case of the triangular lattice heisenberg antiferromagnet with bond disorder. However, the treatment is in principle easily extensible to other spin-models with classical non-collinear ordering. We first start by considering the effect of an isolated bond impurity. On a non-collinear order this immediately leads to local reduction of frustration amongst the spins attached to the bond and therefore, as a result, spins reorient. A simple spin-wave expansion around the 120° classically ordered state of the triangular lattice heisenberg antiferromagnet with an isolated bond impurity leads to the addition of a local perturbing term in the hamiltonian.

$$\delta H_i = wc \left(\frac{\delta J \mathbf{e}_\delta}{J \sqrt{S} a} \right) \cdot \nabla S_i^\perp \quad (11)$$

Where $c = JSa^2$ is the spin-wave velocity, w is a numerical prefactor, $\mathbf{e}_\delta$ is an unit vector directed across the defective bond and $S^\perp$ is the transverse component of the spin in the body frame of the original spin configuration and projected onto the ordering plane. The three goldstone modes of the problem emerges once we expand the term around the three momenta, $Q = (0, 0)$ and $Q = (\pm 4\pi/3, 0)$. It is easy to verify that this particular impurity term is most strongly coupled to the in-plane $Q = (0, 0)$ goldstone mode. If a spatial distribution of such impurities are considered we arrive at the scenario of bond disorder. With the effect of a single impurity clearly understood we can immediately write down a simple free goldstone mode action for the disordered case.

$$S = \int d\tau \int d^d x \rho \left[\frac{1}{2} (\partial_\mu \varphi)^2 + \mathbf{h} \cdot \nabla \varphi \right] \quad (12)$$

Here $\mathbf{h}(\mathbf{x})$ are three-component uncorrelated random fields, $\overline{\mathbf{h}(\mathbf{x})} = 0$ and $\overline{h_i(\mathbf{x})h_j(\mathbf{x}')} = \Delta^2 \delta_{ij} \delta^{(d)}(\mathbf{x} - \mathbf{x}')$. The effect of bond disorder is therefore to introduce a random field disorder which couples to one of the goldstone modes of the symmetry broken phase. Another point to note is that the overall sign of this additional term is fixed by the chirality symmetry breaking of the ground state. The breakdown of time reversal symmetry, chirality and the global spin rotation symmetry all conspires together to generate this random dipolar term. Given, we have a theory of non-interacting bosons coupled to a random dipolar field, we are still in a position to treat the effect of disorder exactly. In momentum space the action takes up the following form.

$$S = \rho \int \frac{d\omega}{2\pi} \int \frac{d^d k}{(2\pi)^d} \left[\frac{1}{2} \varphi(-\omega, -\mathbf{k}) (\omega^2 + \mathbf{k}^2) \varphi(\omega, \mathbf{k}) + 2\pi i \delta(\omega) \frac{\mathbf{h}(-\mathbf{k}) \cdot \mathbf{k} \varphi(\omega, \mathbf{k}) - \mathbf{h}(\mathbf{k}) \cdot \mathbf{k} \varphi(-\omega, -\mathbf{k})}{2} \right] \quad (13)$$

It is easily brought in to a gaussian form by performing the transformation,

$$\varphi(\omega, \mathbf{k}) = \eta(\omega, \mathbf{k}) - 2\pi i \delta(\omega) \frac{\mathbf{h}(-\mathbf{k}) \cdot \mathbf{k}}{\omega^2 + \mathbf{k}^2} \quad (14)$$

with the action sans an unimportant constant given by, $S = \rho \int \frac{d\omega}{2\pi} \int \frac{d^d k}{(2\pi)^d} \eta(-\omega, -\mathbf{k}) (\omega^2 + \mathbf{k}^2) \eta(\omega, \mathbf{k})$.

The disorder averaged fluctuation for the goldstone modes therefore reads,

$$\begin{aligned} \overline{\langle \varphi(\tau, \mathbf{x}) \varphi(\tau', \mathbf{x}') \rangle} &= \int \frac{d\omega}{2\pi} \int \frac{d^d k}{(2\pi)^d} e^{-i(\omega(\tau-\tau') + \mathbf{k} \cdot (\mathbf{x}-\mathbf{x}'))} \langle \eta(-\omega, -\mathbf{k}) \eta(\omega, \mathbf{k}) \rangle \\ &\quad - \Delta^2 \int \frac{d^d k}{(2\pi)^d} \frac{1}{\mathbf{k}^2} e^{-i(\mathbf{k} \cdot (\mathbf{x}-\mathbf{x}'))} \\ &= \int \frac{d^D k}{(2\pi)^D} e^{-i(k(x-x'))} \frac{1}{k^2} - \Delta^2 \int \frac{d^d k}{(2\pi)^d} \frac{1}{\mathbf{k}^2} e^{-i(\mathbf{k} \cdot (\mathbf{x}-\mathbf{x}'))} \end{aligned} \quad (15)$$

with $D = d + 1$. Evidently, we see that for non-zero disorder strength Δ , the correlation function has an infrared divergence at $d = 2$. This would imply that the goldstone modes associated with the non-collinearly ordered triangular lattice antiferromagnet is unstable against even very small bond disorder at $d = 2$. In other words, below or at the critical dimension $d = 2$ bond disorder prevents spontaneous

symmetry breaking in this particular antiferromagnet at $T = 0$.

Although this argument for bond disorder destroying long range order is very robust, it is indeed based on the fluctuation estimates for free goldstone bosons. However, interactions are important ingredients for an effective theory at low dimensions and they may in principle renormalize the disorder coupling of the problem to irrelevancy. Therefore, in the following we reconsider the interacting goldstone boson theory for the ground state of a non-collinear magnet and this time extend it to include the effects of random exchange couplings.

3. *Interacting goldstone modes coupled to a random dipolar field*

In order to find the appropriate goldstone mode action in the presence of bond-disorder in a non-collinear magnet we have to go back to the microscopic expression for such a term. Considering an isolated impurity within one of the elementary plaquettes and between two particular sublattices we find that the additional contribution from a bond impurity $J_{AB} = J + \delta J$ is given by,

$$\delta H = \delta J \mathbf{S}_A(\mathbf{x}_A) \cdot \mathbf{S}_B(\mathbf{x}_B) \quad (16)$$

. Unlike the scenario involving the bulk, the first non-trivial energy contribution from goldstone mode excitations would now come from the single gradient term along the directed unit vector $\mathbf{u}_{AB}$ along the bond. Like before, we use the convenient orthonormal basis transformation and find the impurity contribution to be of the following form.

$$\begin{aligned} \delta H &= \delta J S^2 a \mathbf{n}_A U^{-1} \cdot (\mathbf{u}_{AB} \cdot \nabla) U \mathbf{n}_B \\ &= -\frac{\delta J S^2 a}{\sqrt{12}} (\mathbf{e}_2 U^{-1} \cdot (\mathbf{u}_{AB} \cdot \nabla) U \mathbf{e}_1 - \mathbf{e}_1 U^{-1} \cdot (\mathbf{u}_{AB} \cdot \nabla) U \mathbf{e}_2) \end{aligned} \quad (17)$$

where we have exploited the orthonormality of the basis to drop the terms $\mathbf{e}_1 U^{-1} \nabla U \mathbf{e}_1$ and $\mathbf{e}_2 U^{-1} \nabla U \mathbf{e}_2$ which are zero. Similar considerations for bond impurity across the other two bond direction reveals that we have the same form of the impurity coupling once expressed in terms of the orthonormal basis. Now we

introduce a local dipolar field, $\mathbf{h} = \frac{\delta J}{\sqrt{3}Ja} \mathbf{u}_{AB}$ and the impurity term assumes the form,

$$\begin{aligned} \delta H &= \mathbf{h} \cdot \text{Tr} \left(P t^3 \begin{pmatrix} \mathbf{e}_1^T \\ \mathbf{e}_2^T \\ \mathbf{e}_3^T \end{pmatrix} U^{-1} \nabla U(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \right) \\ &= \mathbf{h} \cdot \text{Tr} (P t^3 U^{-1} \nabla U) \end{aligned} \quad (18)$$

where the same gauge choice from the bulk action was adopted and it is to be noticed that the $SO(3)$ generator corresponding to in plane rotation $t^3 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ makes an appearance. The latter has to do with the fact that in the triangular lattice the local dipolar field arising due to bond disorder couples directly with the goldstone mode associated with the in plane rotation. Now, it is straightforward to generalize to the case of spatially distributed bond impurities i.e. disordered bonds. For a generic scenario involving the emergent random dipolar field coupled with multiple goldstone modes the hamiltonian shall take up the following form.

$$H = \int d^d x \left[\text{Tr} (P (U^{-1} \nabla U)^2) + \mathbf{h} \cdot \text{Tr} (P (c_a t^a) (U^{-1} \nabla U)) \right] \quad (19)$$

As before the interacting nature of the goldstone modes are apparent once we perform an expansion of the form $U^{-1} \partial_\mu U$ in terms of the G/H generators with $U = e^{\xi \cdot \mathbf{t}}$.

$$U^{-1} \partial_\mu U = t^a \omega_a^\mu(\xi) = t^a \left[\partial_\mu \xi_a - \frac{1}{2} f_{abc} \xi_b \partial_\mu \xi_c + \frac{1}{6} f_{abc} f_{cde} \xi_b \xi_d \partial_\mu \xi_e + O(\xi^4) \right] \quad (20)$$

Since, we shall not be concerned with the critical point of the ordered phase we can ignore the differences between various spin-wave velocities and obtain a simpler goldstone mode action.

$$S = \frac{1}{2} \int d\tau \int d^d x \left[\rho_a \omega_a^\mu(\tau, \mathbf{x}) \omega_a^\mu(\tau, \mathbf{x}) + c_a \rho_a h_i(\mathbf{x}) \omega_a^i(\tau, \mathbf{x}) \right] \quad (21)$$

where we used the normalization $\text{Tr}(P t^a t^b) = (\rho_a/2) \delta^{ab}$ and f_{abc} are the structure constants of the broken symmetry G/H ,

$$[t^a, t^b] = f_{abc} t^c \quad (22)$$

. In the case of triangular lattice we have $G/H = SO(3)$ so $f_{abc} = \epsilon_{abc}$, i.e. the levi-civita tensor.

With random bond disorder we may consider a gaussian distribution for the inhomogeneous dipolar fields,

$$P[\mathbf{h}] = \int \mathcal{D}\mathbf{h} e^{-\frac{1}{2\Delta^2} \int d^d x \mathbf{h}(\mathbf{x})^2} \quad (23)$$

and this leads to the homogeneous replica action of our interest.

$$S = \frac{1}{2} \left[\rho_a \sum_r \int d\tau \int d^d x \omega_a^{r,\mu}(\tau, \mathbf{x}) \omega_a^{r,\mu}(\tau, \mathbf{x}) \right. \\ \left. - c_a^2 \rho_a^2 \Delta^2 \sum_{r,s} \int d\tau \int d\tau' \int d^d x \omega_a^{r,i}(\tau, \mathbf{x}) \omega_a^{s,i}(\tau', \mathbf{x}) \right] \quad (24)$$

There are two different types of couplings of the problem, ρ_a and Δ and their mass dimensions are the following.

$$[\rho_a] = d - 1 \\ [\Delta] = (2 - d)/2 \quad (25)$$

This immediately tells us that disorder is marginal in $d = 2$ and relevant at $d < 2$. This result is consistent with the dimensionality of the infrared catastrophe related to the goldstone mode fluctuation. However, at the same time we face the problem that our homogeneous couplings and the disorder coupling do not have a well defined perturbative expansion at the same spatial dimension for $T = 0$. Around $d = 1$ the homogeneous couplings ρ_a are marginal but the disorder coupling Δ is strongly relevant and flows to infinity. Unfortunately this does not tell us anything about what happens close to spatial dimension $d = 2$ when the disorder is marginal. Therefore, in the following we consider the alternate scenario with an expansion at $d = 2 - \epsilon$. The homogeneous coupling is relevant now but we shall only be concerned with the independent flow of the disorder coupling.

4. Renormalization of the disorder coupling Δ at $d = 2 - \epsilon$

[1] Imry and Ma, 75

[2] Mermin and Wagner, 66

[3] Aharony, 78

[4] Azaria, Delamotte and Mouhanna, 92

- [5] Azaria, Delamotte, Delduc and Jolicoeur, 93
- [6] Dombre and Read, 88